

Introduction to Linear and Matrix Algebra

Personal Notes

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1 Vectors in 2D

Linear algebra provides a common language for geometry and data. It describes distances, angles, projections, rotations, and other transformations in any dimension. Computer graphics uses these operations to move a three-dimensional scene and project it onto a two-dimensional display. The original PageRank method instead encoded links among web pages in a matrix and used eigenvectors to assign importance recursively. Both applications reduce to operations on the same basic objects.

1.1 Vectors as Displacements

Definition 1.1 (TWO-DIMENSIONAL VECTOR). A *two-dimensional vector* is an ordered pair of real numbers

$$\mathbf{v} = (v_1, v_2) \in \mathbb{R}^2.$$

The entries v_1 and v_2 are the vector's horizontal and vertical components, respectively.

Vectors may be written as $\mathbf{v}$, $\vec{v}$, or a directed segment such as $\overrightarrow{AB}$. The last notation emphasizes a displacement from A to B .

Geometrically, $\mathbf{v} = (v_1, v_2)$ is an arrow that moves v_1 units horizontally and v_2 units vertically. Its entries do not specify a location: every arrow that moves three units right and two units up represents $(3, 2)$.

Definition: A vector is in **STANDARD POSITION** when its tail is at the origin. The head of (v_1, v_2) is then the point with coordinates (v_1, v_2) .

***Note:** Points describe locations; vectors describe displacements. A point and a vector may use the same pair of coordinates, but their geometric roles are different.*

The order of the entries matters. For example, $(2, 3)$ moves two units horizontally and three vertically, while $(3, 2)$ makes the opposite assignment. These vectors are therefore unequal.

Definition 1.2 (EQUALITY AND THE ZERO VECTOR). Two vectors are equal exactly when their corresponding entries are equal:

$$(v_1, v_2) = (w_1, w_2) \iff v_1 = w_1 \text{ and } v_2 = w_2.$$

Equivalently, equal vectors have the same length and direction. The **ZERO VECTOR** is $\mathbf{0} = (0, 0)$; it represents no displacement.

The symbol $\mathbb{R}^2$ denotes the set of all ordered pairs of real numbers, hence the set of all two-dimensional real vectors.

1.2 Vector Addition

To add vectors, place the tail of the second at the head of the first. The sum is the total displacement, found algebraically by adding corresponding entries:

$$\mathbf{v} + \mathbf{w} = (v_1, v_2) + (w_1, w_2) = (v_1 + w_1, v_2 + w_2).$$

Example: Let $\mathbf{v} = (3, 2)$ and $\mathbf{w} = (1, 3)$. Following $\mathbf{v}$ and then $\mathbf{w}$ produces a total horizontal displacement of $3 + 1 = 4$ and a total vertical displacement of $2 + 3 = 5$. Therefore

$$\mathbf{v} + \mathbf{w} = (4, 5).$$

This head-to-tail construction connects the algebraic and geometric views: componentwise addition records the total displacement along a two-step path.

1.3 Scalar Multiplication

A **SCALAR** is a real number. Multiplying by a scalar changes a vector's magnitude and reverses its direction when the scalar is negative.

Definition 1.3 (SCALAR MULTIPLICATION). For $c \in \mathbb{R}$ and $\mathbf{v} = (v_1, v_2) \in \mathbb{R}^2$, define

$$c\mathbf{v} = (cv_1, cv_2).$$

If $c > 0$, the direction is unchanged; if $c < 0$, it is reversed; and if $c = 0$, the result is $\mathbf{0}$.

For example, if $\mathbf{v} = (2, 1)$, then

$$2\mathbf{v} = (4, 2), \quad -\mathbf{v} = (-2, -1), \quad -\frac{1}{2}\mathbf{v} = (-1, -\frac{1}{2}).$$

All three vectors lie on the same line through the origin; the factor controls their extent and direction along that line.

1.4 Vector Subtraction

Subtraction combines addition with multiplication by -1 :

$$\mathbf{v} - \mathbf{w} = \mathbf{v} + (-1)\mathbf{w} = (v_1 - w_1, v_2 - w_2).$$

When $\mathbf{v}$ and $\mathbf{w}$ are both in standard position, $\mathbf{v} - \mathbf{w}$ points from the head of $\mathbf{w}$ to the head of $\mathbf{v}$. Indeed,

$$\mathbf{w} + (\mathbf{v} - \mathbf{w}) = \mathbf{v},$$

so following $\mathbf{w}$ and then $\mathbf{v} - \mathbf{w}$ ends at the head of $\mathbf{v}$.

Example: For $\mathbf{v} = (3, 1)$ and $\mathbf{w} = (2, 3)$,

$$\mathbf{v} - \mathbf{w} = (3 - 2, 1 - 3) = (1, -2).$$

Starting at the head of $\mathbf{w}$, this displacement moves one unit right and two units down to reach the head of $\mathbf{v}$.

1.5 Symmetry in a Regular Hexagon

Let $\mathbf{u}_1, \dots, \mathbf{u}_6$ point from the center of a regular hexagon to its vertices. Computing every coordinate with trigonometry is unnecessary: symmetry arranges the vectors into three opposite pairs $\mathbf{u}$ and $-\mathbf{u}$. Each pair sums to zero, so

$$\mathbf{u}_1 + \dots + \mathbf{u}_6 = \mathbf{0}.$$

The same reasoning works for any centrally symmetric arrangement of vectors: opposite pairs cancel without requiring their individual coordinates.

Note: Linear algebra problems often admit both an algebraic solution using entries and a geometric solution using direction, symmetry, or shape. Keeping both views available can replace a long computation with a structural argument.

1.6 The Parallelogram Rule

Place nonparallel vectors $\mathbf{v}$ and $\mathbf{w}$ in standard position as adjacent sides of a parallelogram. Its diagonals have direct algebraic meanings:

- $\mathbf{v} + \mathbf{w}$ is the diagonal from the common tail to the opposite corner;
- $\mathbf{v} - \mathbf{w}$ is the diagonal from the head of $\mathbf{w}$ to the head of $\mathbf{v}$.

This rule packages the head-to-tail interpretation of addition and the head-to-head interpretation of subtraction into one geometric picture.

Source: Nathaniel Johnston, Mount Allison University MATH 2221, Lecture 1, "Vectors in 2D," [official video](#) and [course materials for Introduction to Linear and Matrix Algebra](#).